esa ECOSPHERE

Directional associational plant defense from Red deer (*Cervus elaphus*) foraging decisions

Ulrika A. Bergvall† and Olof Leimar

Department of Zoology, Stockholm University, SE-106 91 Stockholm, Sweden

Citation: Bergvall, U. A., and O. Leimar. 2017. Directional associational plant defense from Red deer (*Cervus elaphus*) foraging decisions. Ecosphere 8(3):e01714. 10.1002/ecs2.1714

Abstract. The concept of associational plant defenses is widely accepted and implies that an unpalatable plant can protect its neighbors from grazing. We have investigated the new but similar question of whether a part of a plant, for example, the top or bottom, can protect other parts. At the same time, we investigated whether the previously observed selection of the apical shoot and upper leaves of plants is a direct consequence of food quality (the plant vigor hypothesis) or whether there is an innate or learnt foraging pattern behind the observation. In experiments, we used 1 m high artificial trees, made from aspen branches, and measured red deer browsing from the top (above 0.5 m) and bottom (below 0.5 m), with application of condensed tannin to the top or bottom as a proxy for plant part unpalatability. There were four treatments where either none, both, or one part (top or bottom) of the artificial trees had tannin applied. As expected, we found that red deer consumed less from parts with tannin. We also found that a defended top protected an undefended bottom, but we found no evidence for the opposite relationship, which could be explained by foraging behavior. When examining the behavior, we found that adult red deer prefer to start feeding from the top of a plant. We also found that they spent a shorter time feeding on a defended top. This behavior might cause a defended top to protect an undefended bottom. Such directional associational plant defense could be the result of selectivity with limited flexibility and might be more pronounced for mammalian herbivores than for insects, since mammals are bigger in size and more restricted in their head position. An important applied aspect of these results is that when saplings are protected by adding a repellent, for instance in forestry, it might be enough to apply repellent to the tops. On the other hand, according to this directional associational plant defense, protected bottoms will not protect tops, so newly grown apical shoots may need new protection.

Key words: apical shoot; food selection; plant secondary compounds; sapling; tannin.

Received 20 January 2017; accepted 24 January 2017. Corresponding Editor: Debra P. C. Peters.

Copyright: © 2017 Bergvall and Leimar. This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

† **E-mail:** ulrika.alm-bergvall@zoologi.su.se

Introduction

Herbivores' preferences for certain plants or parts of a plant can be important for forestry and silviculture (Bergqvist et al. 2001), as well as for the management of open land, when the aim is to retard reforestation (Putman and Moore 1998, Gill and Beardall 2001). A good understanding of the foraging decisions of herbivores under different conditions helps to predict herbivore impact. On a larger spatial scale, tree species diversity and stand structure are important factors for foraging decisions (Kuijper 2011). At small and medium scales, plant height, shape, number of lateral branches and defense substances, as well as plant density and distribution, are among the factors that can influence herbivore consumption (McNaughton 1978, Rhoades 1979, Iason et al. 1996, Augustine and McNaughton 1998, Renaud et al. 2003). The consumption

can also be related to the context in which the plant occurs. This is known as associational plant defense, when a target plant gets protection from surrounding plants, or as associational plant susceptibility, when a target plant is more consumed by its association with palatable plants (Atsatt and O'Dowd 1976, Milchunas and Noy-Meir 2002).

An important example of small-scale herbivore selectivity is the tendency to attack the apical shoots and upper leaves of small trees and saplings to a greater extent than other parts (Price 1991). This consumption pattern has been observed in insects as well as in mammalian herbivores (Price 1991, Bergqvist et al. 2001). A possible reason is that the apical shoot and upper leaves are the most actively growing parts, with lower amounts of plant secondary compounds, and thus of high quality, which is in line with the plant vigor hypothesis (Price 1991). Consumption of the apical shoot can lead to multiple trunks and reduced growth (Melzer 1974, Eiberle 1979, Welch et al. 1992), which is costly for the forestry industry (Bergqvist et al. 2001). In addition, already-browsed trees produce leaves with less toxins and more nutrients, which can result in re-browsing of already-damaged trees (Danell et al. 1985, Moore et al. 2000). This susceptibility has been described in several studies as the opposite to a rapidly induced defense (reviewed in Nykänen and Koricheva 2004). Such observations describe the outcome of the foraging behavior, but not the underlying mechanism, and the reason for the preference for apical shoots is still not clear. One explanation could be that there is an innate or learnt foraging pattern where animals start to eat from the top, and if they reach less palatable parts of a tree, they change to another tree. Such behavior could generate the observed pattern and is not incompatible with the plant vigor hypothesis; rather, variation in plant vigor could be the ultimate reason for an innate foraging pattern, or for the learning of the foraging pattern early in life. However, if animals consistently start to eat from the top, this could result in within-plant associational defense.

We thus propose that the principle of associational defense could be applied to situations where different parts of a plant differ in defense, such as when tops of saplings are more palatable than lower parts. Thus, a part of a plant, for example, top or bottom, could protect other parts, or instead, make other parts more susceptible. If a herbivore is unsystematic in where it starts to eat from a small tree, associational protection would be independent of whether the top or the bottom of the tree is less palatable. If, on the other hand, the animal consistently starts to eat from the tops, only bottoms could benefit from within-plant associational defense, and this would be a directional associational defense. Such foraging patterns are of general interest for the understanding of foraging behavior. There are also applied reasons to investigate directional associational plant defense, for instance when small trees are protected by the application of a repellent compound.

Forestry is moving toward management of multiple values, with a recognition of non-timber values in sustainable forest management, including ecological, aesthetic, and recreational values. From this follows a challenge to combine high production (Nordlund and Westin 2011) with other ecosystem services and ecological values (Gamfeldt et al. 2013). One way to facilitate the coexistence of wildlife and forestry is by the artificial protection of saplings. There have been several attempts to redirect browsing from the apical shoot to other parts of the tree or to other trees (Andelt et al. 1992, Nolte 1998). The food quality, and thus preference, is the combined effect of the nutrient content, plant secondary compounds, and the needs and detoxification abilities at a particular point in time (Kimball and Provenza 2003). As a consequence, a repellent must have a physiological effect (antinutritional or toxic) and an associated signal to be efficient in the long run (Kimball et al. 2009). From studies of natural feeding behavior of red deer, it has been shown that they consume less tannin than the average in available foods, implying that they are selecting to avoid tannin (Verheyden-Tixier et al. 2008). Red deer take bites from branches between 16 and 235 cm above ground when they browse (Nichols et al. 2015), and observational studies indicate that they prefer to browse at shoulder height, about 85 cm above ground (Renaud et al. 2003).

Here, we report on an experiment with red deer, designed to distinguish between two possible explanations for the previously observed herbivore selection of apical shoots and upper

leaves: The phenomenon could (1) be an immediate effect of food quality (plant vigor hypothesis) or (2) be an innate or learnt foraging pattern of starting to feed from the top. At the same time, we investigated whether there is within-plant associational defense and if so, whether the defense is directional. In the experiment, we used 1-m artificial trees and measured deer browsing from the top (0.5–1 m) and the bottom (0.5 m–ground). There were four treatments where either none, both, or one part, top or bottom, of the artificial trees had tannin applied. We also observed from what height an animal started to eat, and the duration of each foraging bout depending on food quality.

Materials and Methods

Study species and experimental setup

We used 11 red deer (five females, one male, and five fawns), freely foraging in a 12-ha enclosure with meadows and trees at Tovetorp Zoological Research Station, south central Sweden. The deer had ad libitum access to pasture, water, salt, and mineral stone and were fed with 0.2 kg of food intended for wild cervids (Renfor, Lantmännen) every day. The study took place during June and July 2015 and was performed with permission from the Swedish National Board for Laboratory Animals (Dnr 14-14).

For the experiment, we used fresh aspen (*Populus tremula*) branches that were taken from trees in the enclosure every day approximately 1.5 h before the experiment. Each branch was cut so it reached 1 m above ground and a mark was made at 0.5 m in order to separate the top from the bottom part. The branches were arranged in the experimental area by tying them to 16 plastic fence posts (1.2 m) placed in a 4 × 4 square with 1 m between posts. The deer had previously been habituated to the posts. As a proxy for plant part unpalatability, we applied tannin to the branches. In order to control for differences between parts of the branches other than the tannin application, we mounted every second branch upside down on the post. The part above 0.5 m is referred to as the top, independently of how the branch was mounted. The number of shoots on each branch part was counted before and after each trial.

The experiment had four treatments where none, both, or one part (Top or Bottom) of each branch in the square had tannin applied. There were four trials per treatment, making a total of 16 trials, arranged such that each treatment occurred once in every block of four consecutive trials. There was at most one trial per day. For tannin application, we used a 15% water solution of condensed tannin (Quebracho powder, Unitan Ato containing 75–77% condensed tannin). The solution was applied by soaking the branch or branch part in the solution and letting it dry (see Bergvall et al. 2008). No trials were performed when it was raining. Two motion-triggered remote cameras (Scout Guard 550) were set up to film and take pictures of the red deer during foraging. We measured the number of times and for how long the deer visited the experimental area. In addition, we noted whether animals (divided into adults and fawns) started to eat from the top or the bottom. We also measured the duration of each foraging bout per plant part, tannin application, and animal age.

We used an experimental design where the trial was set as the replicate. This approach has mainly been used when the experimenter is restricted to use the same group of individuals and does not have the possibility to handle the animals individually. Typically, a group of captive animals are repeatedly exposed to a set of different treatments to investigate the relative difference between treatments (moose: Felton et al. 2016; deer and moose: Shipley and Spalinger 1995; cow: Utsumi et al. 2009). Alternatively, a group of wild or semi-wild animals are exposed to a set of different treatments, but the experimenter does not always know the number of animals or the point in time when the animals visit the different treatments and replicates (moose: Danell et al. 1991, Rea et al. 2010).

Statistics

We recorded the consumption (number of eaten shoots) from each part (Bottom and Top) of the 16 branches in a trial. Each trial then gave rise to 32 measurements of consumption, and we used trial as a random effect in mixed-effect statistical models. The fixed effects were the four tannin application treatments (All applied, Top applied, Bottom applied, None applied; referred to as A, T, B, N), the vertical level of the measurement (Bottom, Top), and the position of the branch in the square (Edge, Middle). To aid the

Table 1. Summary of the fixed effects in a logistic mixed-effect model of the probability of herbivory on Bottom and Top parts of aspen branches.

Effect	Estimate	Standard error	*z*-Value	*P*
Intercept	0.34	0.67	0.51	0.61
Position Middle	−0.56	0.32	−1.77	0.08
C1. Bottom vs. Top	−0.16	0.15	−1.07	0.28
C2. Bottom: A and B vs. N and T	−0.84	0.68	−1.24	0.22
C3. Bottom: A vs. B	−0.59	0.89	−0.66	0.51
C4. Bottom: N vs. T	2.84	1.01	2.61	0.009
C5. Top: A and T vs. B and N	−2.31	0.71	−3.25	0.001
C6. Top: A vs. T	0.34	0.96	0.36	0.72
C7. Top: B vs. N	−0.49	1.00	−0.49	0.62

Notes: The intercept represents the average probability of herbivory on Edge branches. The tannin application treatments were A (All applied), T (Top applied), B (Bottom applied), and N (None applied). The effects of this treatment factor and the vertical level of herbivory (Bottom/Top) are expressed as seven orthogonal contrasts (C1 to C7). The effects are given on the logit scale.

interpretation, we performed the analysis of the effects of treatment and vertical level as planned comparisons, using seven orthogonal contrasts involving these factors. We based the contrasts on the principle of comparing two groups of observations, and then comparing two groups within each of these groups, and so on, which results in orthogonal contrasts. The seven comparisons are labeled C1 to C7 and are described in Table 1. Of particular relevance to our hypotheses are comparison C3 and C4—which, respectively, test whether there are associational susceptibility and defense of the Bottom part from the Top part—and comparison C6 and C7—which, respectively, test whether there are associational susceptibility and defense of the Top part from the Bottom part. In addition, C2 tests whether own defense is important for Bottom parts, and C5 tests whether own defense is important for Top parts.

As response variable, we first tried the proportion of consumed shoots on a branch part, using the lmer function in the lme4 package (Bates et al. 2015) in the R statistics package (R Core Team 2015). The proportions had a bimodal distribution, with modes at 0 and near 1, and we did not manage to achieve normally distributed residuals through transformation of the response variable, although the transformation log(proportion + 0.1) resulted in homogenous variances. As an alternative, we used whether a branch part suffered consumption or not (proportion positive or zero) as a response variable and fitted a logistic generalized mixed-effect model, using the glmer function in the lme4 package.

To test whether the animals started to eat from the tops or the bottoms, we performed a chi-squared goodness-of-fit test to equal proportions of starts from top and bottom. We examined all foraging events, including those with investigation of leaves without eating, as well as just the foraging events where leaves were consumed, for adults and fawns separately. In addition, we tested whether the time an adult animal ate from the top of a branch was related to tannin application. For this, we used Mann–Whitney *U* test, because the times were not normally distributed.

Results

From the statistical analysis of the herbivory on branch parts (Table 1), there was a non-significant tendency for a higher risk of herbivory for branches at the edge position. For the seven orthogonal contrasts, the first comparison (contrast C1 in Table 1) showed that bottoms and tops did not differ significantly in the overall probability of herbivory. The second comparison (C2 in Table 1) showed that own defense did not have a statistically significant influence on the risk of herbivory on bottom parts, whereas for tops, own defense reduced the risk (C5 in Table 1). There were no statistically significant effects of within-branch associational susceptibility (C3 and C6 in Table 1). Interestingly, there was a within-branch associational defense of bottoms from defended tops (C4 in Table 1), but no corresponding statistically significant defense of tops from defended bottoms (C7 in Table 1).

Each part (top or bottom) of the branches consisted of 6.4 ± 2.2 shoots (mean ± SD; N = 512), and each shoot consisted of 5.9 ± 0.63 leaves (mean ± SD; N = 36). Fig. 1A displays the data as averages of the proportion consumed shoots and agrees with the analysis in Table 1, in that a defended top protected an undefended bottom, but not the opposite relationship. However, the data on proportion consumed in Fig. 1A give a different picture than the analysis based on the risk of herbivory (Table 1), in that the overall

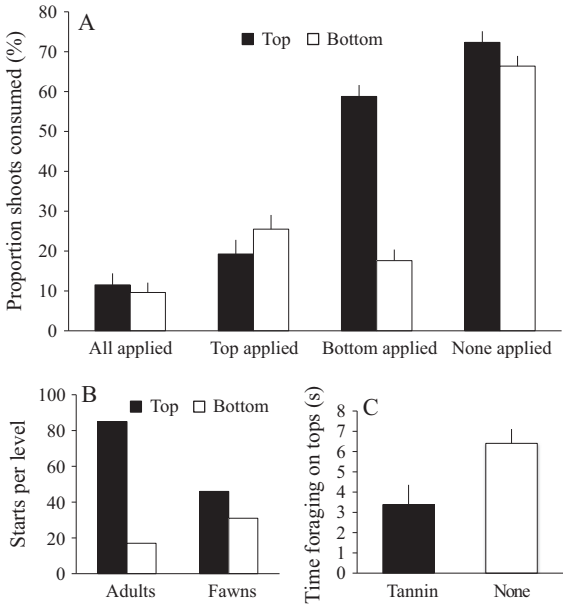

Fig. 1. (A) The proportion browsed shoots from branch parts in relation to tannin application treatment and vertical level (top or bottom) in the foraging experiment with red deer. Sixteen aspen branches were treated with a 15% tannin solution at the top or the bottom parts, none, or all parts. (B) The number of times an adult or fawn started to eat from the top or the bottom of the trees. (C) The time an adult red deer spent foraging on a top dependent on its application of tannin (n_{tannin} = 21, n_{none} = 64). Data are given as mean + SE.

proportion consumed from tops appears higher than that from bottoms (C1), and that own defense appears to reduce the proportion consumed from bottoms (C2). Both these effects were supported by the (possibly unreliable) model fitting using lmer on log-transformed proportions ($P < 0.001$ and $P = 0.04$, respectively). In any case, both for the proportion consumed and the probability of herbivory, the consumption pattern generated a directional associational defense from top to bottom.

The adult deer generally started to eat from the tops, but fawns did not show any statistically significant preference for starting from tops over bottoms (Table 2; Fig. 1B). When comparing starts from tops, we found that an adult deer starts more often from a top in the Bottom applied treatment than in the Top applied treatment (Table 3). In general, fawns showed a different consumption pattern from adults, with slower feeding and less consumption. The adult feeding pattern of starting from the tops held both if we looked at all foraging events or just those where consumption occurred (Table 2). When an adult started feeding from a top, the time feeding was shorter if tannin had been applied to it than without tannin application (Fig. 1C; Mann–Whitney U test; n_{tannin} = 21, n_{none} = 64, U = 436, Z = 2.40, P = 0.02), and this behavior might contribute to the pattern of within-branch associational defense.

The red deer visited the experimental area on average three times a day, for a total of 11 min. The group usually visited the area at the same time, but not all deer ate at the same time. The deer spent longer time in the experimental area in the N treatment, where no tannin was applied to the branch parts, than in the other treatments (ANOVA: F = 5.00, df = 3, P = 0.02).

Discussion

Our experiment showed that the application of tannin as a defense of the top half of 1 m high artificial trees had a strong effect, both in reducing herbivory on tops and in protecting the bottom half (Fig. 1A, Table 1). Tannin application to the bottom half, on the other hand, did not protect

Table 2. Chi-squared goodness-of-fit test to equal proportions of the number of times red deer started to forage from the top and bottom parts of a branch.

	Bite and nose					Bite				
Animals	Start top	Start bottom	χ^2	P	n	Start top	Start bottom	χ^2	P	n
All	131	48	38.48	<0.0001	179	97	40	23.72	<0.0001	137
Adults	85	17	45.33	<0.0001	102	57	12	29.35	<0.0001	69
Fawns	46	31	2.92	0.09	77	40	28	2.12	0.12	68

Notes: The test on the left-hand side includes all foraging events and on the right-hand side only foraging events where the animal consumed some leaves. The test is shown for all animals and for adults and fawns separately.

Table 3. Chi-squared goodness-of-fit test to equal proportions of the number of times adult red deer started (bite and nose) from the top in the Bottom applied and the Top applied treatments.

Bottom applied	Top applied	χ^2	P	n
16	2	10.88	0.020	18

the top half. Behind this outcome was a foraging pattern where adult red deer predominantly started to eat from the top (Fig. 1B), and consumed less and for shorter time when the top was defended (Fig. 1C). In addition, when we compared the starts from the tops in the Top applied with the Bottom applied, we found that an adult deer started from the top more often in the Bottom applied treatment (Table 3), indicating an avoidance already before consumption. This finding that defended tops protect bottoms, but not the other way around, is a new variant of the idea of associational defense, where unpalatable plants protect their palatable neighbors from browsing. We also tested for effects of within-plant associational susceptibility, but we did not find any.

Associational effects of mammalian herbivory have been interpreted as consequences of selectivity at different hierarchical levels or spatial scales (Danell et al. 1991). An example is associational defense from herbivore selectivity between rather than within patches of plants (McNaughton 1979). From a number of foraging experiments with patchily distributed plants, effects of both within- and between-patch selectivity have been found (e.g., McNaughton 1979, Danell et al. 1991, Hjältén et al. 1993, Bergman et al. 2005, Bergvall et al. 2006, 2008, Rautio et al. 2008, Utsumi et al. 2009, Huang et al. 2016), with factors such as the spatial scale of patches and herbivore size and lifestyle influencing the selectivity. A related perspective, deriving from the study of predator–prey interactions, is to describe the foraging process as sequences of behavior, involving search, detection, and consumption (Stephens and Krebs 1986, Endler 1991). From this perspective, associational defense occurs when the foraging sequence is interrupted at the search stage in a patch of plants, before a focal palatable plant in the patch is detected or consumed (Stutz et al. 2015*a, b*). There is thus directionality in the associational defense, in that the foraging sequence is interrupted before a focal plant is consumed. Our finding that tops protect bottoms of small trees, when herbivores start feeding from the top, is a within-plant example that agrees with this general perspective.

The adult red deer in our study started to eat from above 50 cm from the ground around five times more often than from below 50 cm (Table 2). The fawns started about equally often above as below 50 cm, and their foraging behavior was exploratory with rather little consumption. A simple explanation for these observations is that the starting height is a result of an animal's shoulder height and the typical position of its head. In addition, the height of the attacked tree could also influence the height from which the first bite is taken. A walking deer typically has its head and neck in a position where the mouth is at or slightly lower than the shoulder. The head position can vary with emotional state, age, sex, and type of forage. For instance, a grazing deer walks with a low head position ready for the next bite and can therefore take a bite from a tree from that position, although this is an exception rather than the rule. A few examples can illustrate this suggestion. The average shoulder height of red deer ranges from 110 cm for females to 130 cm for males (Clutton-Brock et al. 1980), so one might expect that they prefer to start to eat from a tree around these heights rather than starting much lower down or higher up. A previous study on adult red deer browsing height found a roughly normal distribution with a mean of about 125 cm above ground, with very few bites higher than 200 cm or lower than 20 cm (Pajtík et al. 2015). Another study on one-year-old red deer ranging from 82 to 105 cm in shoulder height found a mean foraging height of 85 cm above ground (Renaud et al. 2003). The fawns in our study measured about 80 cm above ground in shoulder height, but they held the mouth about 50 cm above ground when they walked toward the artificial trees. Furthermore, roe deer prefer to browse at shoulder height (Duncan et al. 1998), while moose seem to prefer to forage below shoulder height (Nichols et al. 2015). It thus seems like ungulate shoulder height and head position influence the foraging height and perhaps also the height of the first bite, although the distribution of the height of the first bite from a tree has not been directly estimated for red deer, nor for other ungulate species. Overall, these explanations for

the height of the first bite could correspond to an innate foraging pattern, or possibly a learnt foraging style. In general, the size of the browser as well as the trees would be an important factor to take into account when one tries to explain foraging patterns.

The plant vigor hypothesis states that herbivores prefer to feed on young and rapidly growing plants, and applies to both insect and mammal herbivores (Price 1991). At the within-plant level, the hypothesis would explain a preference for apical shoots and upper leaves (Price 1991). Thus, a herbivore that investigates different parts of small trees and saplings, consuming those of highest quality, would mostly browse apical shoots. However, the foraging behavior shown in our study does not support this idea. Instead, we found that defense of the top half of 1 m high trees had an important effect of reducing browsing from both the top and bottom halves (Fig. 1A). On the other hand, undefended tops suffered much browsing (Fig. 1A), so a herbivore preference for higher-quality food is not symmetric between tops and bottoms. As mentioned, our explanation for this is a foraging pattern of starting from the tops for trees of about this height. Results from previous studies on ungulate browsing are sometimes in agreement with the plant vigor hypothesis (Danell et al. 1985, Bergström et al. 2000), but sometimes find that foragers select other than the most vigorously growing parts of trees. Thus, in a study of several African browsers, the highest browsing pressure was found for plant species of intermediate vigor (Makhabu et al. 2006).

Our results could be of use when considering how to influence ungulate foraging decisions by applying repellents to small trees and saplings (Andelt et al. 1992, Nolte 1998, Wagner and Nolte 2001, Santilli et al. 2004). An important challenge for forestry is to combine high production (Nordlund and Westin 2011) with other ecosystem services and ecological values (Gamfeldt et al. 2013), which can include populations of ungulates. Based on our results, it can be enough to apply the repellent to the top of a sapling in order to protect the whole sapling, although this approach might not work for trees above a certain height. On the other hand, when a new apical shoot appears, corresponding to the situation in our experiment with tannin application only to the bottom half, the new shoot will not be protected by repellant from lower down (Fig. 1A). Thus, application of repellent to new shoots is likely to be needed.

In conclusion, our results demonstrated a directional associational plant defense, in the sense that a defended top protected an undefended bottom, but we found no evidence for the opposite relationship. This seems to be the result of selectivity with limited flexibility, in that adult deer typically started foraging from the tops. We suggest that this foraging pattern is a consequence of the size and typical head position of the deer, and might represent an innate foraging style. An applied aspect is that if one attempts to protect saplings by adding a repellent, the important thing is to apply repellent to the tops.

Acknowledgments

This study was supported by the Swedish Research Council FORMAS. We thank Rebecca Stutz for useful discussions.

Literature Cited

Andelt, W. F., D. L. Baker, and K. P. Burnham. 1992. Relative preference of captive cow elk for repellent-treated diets. Journal of Wildlife Management 56:164–173.

Atsatt, P. R., and D. J. O'Dowd. 1976. Plant defense guilds. Science 193:24–29.

Augustine, D. J., and S. J. McNaughton. 1998. Ungulate effect on the functional species composition of plant communities: herbivore selectivity and plant tolerance. Journal of Wildlife Management 62:1165–1183.

Bates, D., M. Meachler, B. Bolker, and S. Walker. 2015. Fitting linear mixed-effects models using lme4. Journal of Statistical Software 67:1–48.

Bergman, M., G. R. Iason, and A. J. Hester. 2005. Feeding patterns by roe deer and rabbits on pine, willow and birch in relation to spatial arrangement. Oikos 109:513–520.

Bergqvist, G., R. Bergström, and L. Edenius. 2001. Patterns of stem damage by moose (*Alces alces*) in young *Pinus sylvestris* stands in Sweden. Scandinavian Journal of Forest Research 16:363–370.

Bergström, R., C. Skarpe, and K. Danell. 2000. Plant responses and herbivory following simulated browsing and stem cutting of *Combretum apiculatum*. Journal of Vegetation Science 11:409–414.

Bergvall, U. A., P. Rautio, K. Kesti, J. Tuomi, and O. Leimar. 2006. Associational effects of plant

defences in relation to within- and between-patch food choice by a mammalian herbivore: neighbour contrast susceptibility and defence. Oecologia 147:253–260.

Bergvall, U. A., P. Rautio, H. Sirén, J. Tuomi, and O. Leimar. 2008. The effect of spatial scale on plant associational defences against mammalian herbivores. Écoscience 15:343–348.

Clutton-Brock, T. H., S. D. Albon, and P. H. Harvey. 1980. Antlers, body size and breeding group size in the Cervidae. Nature 285:565–567.

Danell, K., L. Edenius, and P. Lundberg. 1991. Herbivory and tree stand composition: moose patch use in winter. Ecology 72:1350–1357.

Danell, K., K. Huss-Danell, and R. Bergström. 1985. Interactions between browsing moose and two species of birch in Sweden. Ecology 66:1867–1878.

Duncan, P., H. Tixier, R. R. Hofmann, and M. Lechner-Doll. 1998. Feeding strategies and the physiology of digestion in roe deer. Pages 91–116 *in* R. Andersen, P. Duncan, and J. Linell, editors. The European roe deer: the biology of success. Scandinavian University Press, Oslo, Norway.

Eiberle, K. 1979. Consequences of simulated browsing of roe deer on the development of young forest trees. Schweizerische Zeitschrift für Forstwesen 129:757–768.

Endler, J. A. 1991. Interactions between predators and prey. Behavioural Ecology: An Evolutionary Approach 3:169–196.

Felton, A. M., A. Felton, D. Raubenheimer, S. J. Simpson, S. J. Krizsan, P. O. Hedwall, and C. Stolter. 2016. The nutritional balancing act of a large herbivore: an experiment with captive moose (*Alces alces* L.). PLoS ONE 11:e0150870.

Gamfeldt, L., et al. 2013. Higher levels of multiple ecosystem services are found in forests with more tree species. Nature Communications 4:1340.

Gill, R. M. A., and V. Beardall. 2001. The impact of deer on woodlands: the effects of browsing and seed dispersal on vegetation structure and composition. Forestry 74:209–218.

Hjältén, J., K. Danell, and P. Lundberg. 1993. Herbivore avoidance by association: vole and hare utilization of woody plants. Oikos 68:125–131.

Huang, Y., L. Wang, D.-H. Zeng, and C. Liu. 2016. How does the foraging behaviour of large herbivores cause different associational plant defences? Scientific Reports 6:20561.

Iason, G. R., A. J. Duncan, S. E. Hartley, and B. W. Staines. 1996. Feeding behavior of red deer (*Cervus elaphus*) on Sitka spruce (*Picea sitchensis*): the role of carbon–nutrient balance. Forest Ecology and Management 88:121–129.

Kimball, B. A., and F. D. Provenza. 2003. Chemical defense and mammalian herbivores. In J. D. Hardege, editor. Encyclopedia of life support systems (EOLSS), Developed under the auspices of the UNESCO. EOLSS Publishers, Oxford, UK. http://www.eolss.net

Kimball, B. A., J. Taylor, K. R. Perry, and C. Capelli. 2009. Deer responses to repellent stimuli. Journal of Chemical Ecology 35:1461.

Kuijper, D. P. J. 2011. Lack of natural control mechanisms increases wildlife-forestry conflict in managed temperate European forest systems. European Journal of Forest Research 130:895–909.

Makhabu, S. W., C. Skarpe, and H. Hytteborn. 2006. Elephant impact on shoot distribution on trees and on rebrowsing by smaller browsers. Acta Oecologica 30:136–146.

McNaughton, S. J. 1978. Serengeti ungulates: Feeding selectivity influences the effectiveness of plant defense guilds. Science 199:806–807.

McNaughton, S. J. 1979. Optimization process: grass–ungulate relationships in the Serengeti. American Naturalist 113:691–703.

Melzer, E. W. 1974. The influence of planting age and method on the incidence of browsing by game in young plantations of *Picea abies*. Beiträge für die Forstwirtschaft 8:189–192.

Milchunas, D. G., and I. Noy-Meir. 2002. Grazing refuges, external avoidance of herbivory and plant diversity. Oikos 99:113–130.

Moore, N. P., J. D. Hart, and S. D. Langton. 2000. Browsing by fallow deer (*Dama dama*) in young broadleaved plantations: seasonality, and the effects of previous browsing and bud eruption. Forestry 73:437–445.

Nichols, R., J. P. G. M. Cromsig, and G. Spong. 2015. DNA left on browsed twigs uncovers bite-scale resource use patterns in European ungulates. Oecologia 178:275–284.

Nolte, D. L. 1998. Efficacy of selected repellents to deter deer browsing on conifer seedlings. International Biodeterioration & Biodegradation 42:101–107.

Nordlund, A., and K. Westin. 2011. Forest values and forest management attitudes among private forest owners in Sweden. Forests 30–50.

Nykänen, H., and J. Koricheva. 2004. Damage-induced changes in woody plants and their effects on insect herbivore performance: a meta-analysis. Oikos 104:247–268.

Pajtík, J., B. Konôpka, M. Bošel'a, V. Šebeň, and P. Kaštier. 2015. Modelling forage potential for red deer: a case study in post-disturbance young stands of rowan. Annals of Forest Research 58: 91–107.

Price, W. P. 1991. The plant vigor hypothesis and herbivore attack. Oikos 62:244–251.

Putman, R. J., and N. P. Moore. 1998. Impact of deer in lowland Britain on agriculture, forestry and conservation habitats. Mammal Reviews 28:141–164.

R Core Team. 2015. R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/

Rautio, P., K. Kesti, U. A. Bergvall, J. Tuomi, and O. Leimar. 2008. Spatial scales of foraging in fallow deer: implications for associational effects in plant defences. Acta Oecologica 34:12–20.

Rea, R. V., D. P. Hodder, O. Hjeljord, and A. Langen. 2010. Paper birch (*Betula papyrifera*) shoot selection by moose (*Alces alces*) following a forest-cleaning experiment. Scandinavian Journal of Forest Research 25:157–163.

Renaud, P. C., H. Verheyden-Tixier, and B. Dumont. 2003. Damage to saplings by red deer (*Cervus elaphus*): effect of foliage height and structure. Forest Ecology and Management 181:31–37.

Rhoades, D. F. 1979. Evolution of chemical defence against herbivores. Pages 2–54 *in* G. A. Rosenthal and D. H. Janzen, editors. Herbivores: their interaction with secondary plant metabolites. Academic Press, New York, New York, USA.

Santilli, F., L. Mori, and L. Galardi. 2004. European Journal of Wildlife Research 50:85.

Shipley, L. A., and D. E. Spalinger. 1995. Influence of size and density of browse patches on intake rates and foraging decisions of young moose and white-tailed deer. Oecologia 104:112–121.

Stephens, D. W., and J. R. Krebs. 1986. Foraging theory. Princeton University Press, Princeton, New Jersey, USA.

Stutz, R. S., P. B. Banks, N. Dexter, and C. McArthur. 2015*a*. Herbivore search behavior drives associational plant refuge. Acta Oecologica 67:1–7.

Stutz, R. S., P. B. Banks, N. Dexter, and C. McArthur. 2015*b*. Associational refuge in practice: Can existing vegetation facilitate woodland restoration? Oikos 124:571–580.

Utsumi, S. A., C. A. Cangiano, J. R. Galli, M. B. McEachern, M. W. Demment, and E. A. Laca. 2009. Resource heterogeneity and foraging behavior of cattle across spatial scales. BMC Ecology 9:9.

Verheyden-Tixier, H., P. C. Renaud, N. Morellet, J. Jamot, J. M. Besle, and B. Dumont. 2008. Selection for nutrients by red deer hinds feeding on a mixed forest edge. Oecologia 156:715–726.

Wagner, K. K., and D. L. Nolte. 2001. Comparison of active ingredients and delivery systems in deer repellents. Wildlife Society Bulletin 29:322–330.

Welch, D., D. S. Staines, and D. D. French. 1992. Leader browsing by Red and Roe deer on young Sitka spruce trees in Western Scotland. II. Effects on growth and tree form. Forestry 65:309–330.